

1. Administrative & Study Identification

Full Study Title: A Study to Learn About The COVID-19 (Study) Vaccine (Called COMIRNATY) in People That Are Less Than 21 Years Old **Short Title/Acronym:** CAMP (Low-Interventional Cohort Study of Myocarditis/Pericarditis) **Protocol Number:** C4591036 **NCT Number:** NCT05295290 **Sponsor Name:** Pfizer **Investigational Product:** COMIRNATY (COVID-19 Vaccine, mRNA) **Phase of Development:** Phase IV **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the cardiac and non-cardiac long-term outcomes of persons <21 years of age with myocarditis/pericarditis after the administration of COMIRNATY, compared with similarly aged persons with myocarditis/pericarditis associated with COVID-19, including MIS-C, without exposure to the vaccine. **Secondary Objectives:** To track left ventricular ejection fraction (LVEF), ongoing cardiac dysfunction, arrhythmias, conduction issues, and quality of life scores.

2.2 Background & Rationale To address the need for long-term safety data regarding the risk of myocarditis and pericarditis following COMIRNATY vaccination in the pediatric and young adult populations. This observational study aims to assess whether these post-vaccination cardiac events differ in severity or long-term outcomes compared to the background risk observed following acute COVID-19 infection or Multisystem Inflammatory Syndrome in Children (MIS-C).

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Low-Interventional Cohort Study) **Control Type:** Active Comparator Cohort (COVID-19-related myocarditis/pericarditis, including MIS-C) **Blinding:** Open-label **Randomization:** N/A (Cohort assignment is based on disease etiology and exposure history)

3.2 Planned Enrollment & Duration Target \$N\$: Estimated 400 participants (300 in COMIRNATY cohorts, 100 in COVID-19 comparator cohort) **Number of Sites:** Multicenter **Estimated Study Start:** November 21, 2022 **Estimated Primary Completion:** November 21, 2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age < 21 years (0 to 20 years). 2. Presentation to a participating medical center with evaluation in the Emergency Room and/or hospitalization. 3. Probable or confirmed myocarditis/pericarditis associated with either COMIRNATY (received dose ≤ 7 days of symptom onset) OR acute COVID-19 infection/MIS-C.

4.2 Exclusion Criteria 1. A plausible alternative etiology for myocarditis/pericarditis, as determined by the site based upon routine clinical practice. 2. Pre-existing cardiac conditions that could impact the primary endpoint (e.g., documented history of left ventricular dysfunction, pacemaker, or congenital heart disease). 3. Previous administration with an investigational drug or vaccine within 30 days of enrollment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Observational cohort study; no investigational drug is administered. Participants undergo cardiac imaging (Cardiac MRI, echocardiograms), ECGs, and quality of life surveys. **Route of Administration:** N/A (Observational) **Duration of Treatment:** Long-term follow-up assessments up to 5 years following illness onset.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for the management of myocarditis/pericarditis. **Prohibited:** N/A

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
1	Screening/Baseline	Informed Consent, medical history, initial cardiac evaluation, and cohort assignment (prospective/retrospective).
2	Interim	6 Months Post-Onset
3	End of Study	Up to 5 Years

Primary composite endpoint evaluation: Cardiac MRI, Echocardiogram, ECG, Quality of Life surveys. Final cardiac imaging, vital signs, arrhythmias check, and final long-term quality of life assessments.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: A single composite primary outcome measure defined as the presence of 1 or more of the following 6 months after illness onset: left ventricular dysfunction (LVEF < 55%), findings of myocarditis by Lake Louise criteria on cardiac MRI, or the presence of high-grade

arrhythmia/conduction disturbance. **Measurement Method:** Echocardiogram, Cardiac MRI, and ECG/ambulatory monitoring.

7.2 Secondary & Exploratory Endpoints 1. Arrhythmias and conduction issues up to 5 years post-illness onset. 2. Quality of life (QoL) scores using pediatric tools (BASC-3, PROMIS, PEDS QL) up to 5 years.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous observational tracking of cardiac outcomes, specifically ongoing left ventricular dysfunction and arrhythmias. **Serious Adverse Events (SAEs):** Routine pharmacovigilance and long-term surveillance data reporting via electronic records. **Data Monitoring Committee (DMC):** Managed internally by Pfizer per post-marketing surveillance obligations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Cohort 1 & 2 (COMIRNATY-associated myocarditis) compared against Cohort 3 (Comparator group: COVID-19/MIS-C associated myocarditis). **Sample Size Justification:** Target sample of ~400 participants powered to detect variations in long-term cardiac recovery rates between vaccine-induced and infection-induced inflammation. **Handling of Missing Data:** Standard observational cohort techniques, such as Multiple Imputation (MI) or Last Observation Carried Forward (LOCF) for longitudinal QoL metrics.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and post-marketing observational research guidelines. **Monitoring Plan:** Scheduled remote and onsite data collection from routine clinical practice up to the 5-year follow-up mark. **Audit Trail:** Data verification of imaging central reads, eCRF locking, and adherence to CDC case definitions for myocarditis/pericarditis.

11. Study Identifiers & Governance

Primary ID: NCT05295290 **Registry Number:** NCT05295290 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn About The COVID-19 (Study) Vaccine (Called COMIRNATY) in People That Are Less Than 21 Years Old **Official Regulatory Title:** Low-Interventional Cohort Study of Myocarditis/Pericarditis Associated With COMIRNATY in Persons Less Than 21 Years of Age

12. Clinical Framework (Condition Specific)

Note: Adapted from the template to fit the studied condition.

Primary Condition: Myocarditis / Pericarditis **Diagnosis Threshold:** Probable or confirmed case per CDC definition (elevated troponin, abnormal cardiac function/wall motion, or cMRI findings). **Medical Methodology:** Routine clinical cardiac evaluation and imaging. **Optimization Target:** Resolution of left ventricular dysfunction (LVEF $\geq$ 55%) and absence of arrhythmias.

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for evaluation of potential causes for myocarditis/pericarditis. **Education Protocol (HCP):** N/A (Low-interventional observation) **Education Protocol (Patient):** Completion of standardized pediatric health assessments. **Quality Audit Mechanism:** Central review of cardiac MRIs and echocardiogram data outputs.

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years following illness onset **Onsite Visit Cadence:** Baseline, 6 months (Primary Endpoint), and long-term intervals **Remote Monitoring Frequency:** Intermittent QoL questionnaires (BASC-3, PROMIS, PEDS QL) **Checklist Review Interval:** Associated with follow-up milestones

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				